

Email Analysis Report

A Hope to CoCanDa — Phishing Email Investigation

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Case	A Hope to CoCanDa — BTLO Challenge
Email File	A Hope to CoCanDa.eml
Classification	TLP: WHITE
Verdict	 MALICIOUS — Phishing Email with Spoofed Sender

1. Executive Summary

This report documents the analysis of a suspicious email with the subject "A Hope to CoCanDa" sent to themajornearth@gmail.com. The email was analysed as part of a Blue Team Labs Online (BTLO) challenge to identify phishing indicators and attacker infrastructure.

The email exhibits multiple high-confidence phishing indicators including SPF authentication failure, a mismatched Reply-To address, use of a known malicious mail relay (emkei.cz), and Base64-encoded content containing a suspicious attachment. The sender spoofed the identity of "Bill" using the domain microapple.com — a typosquat likely designed to impersonate Apple Inc.

2. Email Header Analysis

The email headers were extracted from the raw .eml file and examined field-by-field. The following table summarises key header fields and their analysis findings.

Header Field	Value	Status	Analyst Note
Delivered-To	themajornearth@gmail.com	INFO	Intended target recipient
From	"Bill" <billjobs@microapple.com>	SUSPICIOUS	Display name 'Bill' used to impersonate. Domain microapple.com is a typosquat of apple.com
Reply-To	negeja3921@pashter.com	SUSPICIOUS	Differs from From address. Attacker controls this inbox —

Header Field	Value	Status	Analyst Note
Return-Path			replies go to attacker not sender
	<billjobs@microapple.com>	INFO	If delivery fails, bounce sent here. Matches From field in this case
SPF	FAIL	FAIL	93.99.104.210 not authorised to send on behalf of microapple.com — strong spoofing indicator
DKIM	Not present	WARNING	No DKIM signature found — email integrity cannot be verified
DMARC	Not evaluated	WARNING	No DMARC policy enforcement observed
Received (relay)	emkei.cz [93.99.104.210]	MALICIOUS	emkei.cz is a publicly known fake/anonymous email sender service used by threat actors
X-Priority	3 (Normal)	INFO	No urgency manipulation used in headers
Date	Tue, 26 Jan 2021 01:41:18 -0500	WARNING	Date field can be spoofed by sender
Message-ID	<20210126064118.1993E221F8@localhost>	SUSPICIOUS	localhost in Message-ID confirms email was sent from emkei.cz fake mailer, not a legitimate mail server
Content-Type	multipart/mixed	INFO	Indicates multiple content parts — email contains both text body and attachment(s)
Content-Transfer-Encoding	base64	INFO	Email body encoded in Base64 — decoded to reveal actual content

3. Email Authentication Analysis

3.1 SPF (Sender Policy Framework)

SPF check returned FAIL. The sending IP address 93.99.104.210 (emkei.cz) is not listed as an authorised sender in the SPF record for microapple.com. This is a definitive indicator that the email did not originate from a legitimate microapple.com mail server and the sender identity is spoofed.

- Sending IP: 93.99.104.210
- SPF Result: FAIL
- Expected sender domain: microapple.com
- Conclusion: IP not in authorised SPF record — spoofing confirmed

3.2 DKIM (DomainKeys Identified Mail)

No DKIM signature was present in the email headers. DKIM provides cryptographic verification that the email content has not been tampered with in transit. The absence of DKIM means there is no way to verify the integrity of the message body or that it was authorised by the domain owner.

3.3 DMARC

DMARC (Domain-based Message Authentication Reporting and Conformance) was not evaluated or enforced in this case. Without DMARC policy enforcement, emails failing SPF and DKIM checks may still reach recipient inboxes depending on the mail provider's configuration.

4. Sender Infrastructure Analysis

4.1 Sending Server — emkei.cz

The Received header reveals the email was relayed through emkei.cz, a publicly accessible online anonymous email sending service. This platform allows anyone to send emails with any From address without authentication, making it a frequently abused tool by phishing actors and social engineers.

- Server: emkei.cz
- IP Address: 93.99.104.210
- Nature: Anonymous/fake email relay service — known malicious infrastructure
- Threat Actor Usage: Widely used in low-sophistication phishing and BEC attempts
- Recommended Action: Block 93.99.104.210 and emkei.cz at email gateway

4.2 Sender Domain — microapple.com

The sender domain microapple.com is a typosquat domain designed to visually resemble apple.com. This is a classic social engineering technique where a threat actor registers a look-alike domain to deceive victims into believing the communication is from a legitimate, trusted organisation.

- Registered domain: microapple.com
- Target impersonation: Apple Inc. (apple.com)
- Technique: Typo squatting / domain spoofing
- SPF: Not properly configured — permitted attacker to spoof

4.3 Reply-To Address — negeja3921@pashter.com

The Reply-To address negeja3921@pashter.com differs significantly from the From address billjobs@microapple.com. This is a deliberate attacker technique: when a victim replies to the email, the response is routed to the attacker-controlled inbox rather than back to the spoofed sender. The pashter.com domain is a disposable email service commonly abused by threat actors.

- From: billjobs@microapple.com
- Reply-To: negeja3921@pashter.com
- Discrepancy: HIGH — strong phishing indicator
- Purpose: Capture victim replies without exposing real attacker infrastructure

5. Email Content & Attachment Analysis

5.1 Body Content — Base64 Decoding

The email body was encoded using Base64 (Content-Transfer-Encoding: base64). Using CyberChef with the recipe From Base64 → To Hex, the encoded content was decoded. The decoded output revealed the plain text body of the email beginning with:

"Hi TheMajorOnEarth, The abducted CoCanDians a..."

This social engineering lure references fictional colony abduction, crafted to create urgency and curiosity in the target recipient.

5.2 Attachment Analysis — File Signature (Magic Bytes)

The email contained attachments extracted from the PuzzleToCoCanDa directory. File type was verified using magic bytes (file signatures) rather than relying on file extensions alone, as extensions can be easily spoofed.

File Name	Magic Bytes	True File Type	Notes
DaughtersCrown.jpeg	FF D8 FF	JPEG Image	Confirmed JPEG. ExifTool metadata extracted.
GoodJobMajor.pdf	25 50 44 46	PDF File	Magic bytes 25 50 44 46 = %PDF — confirmed PDF.

5.3 ExifTool Metadata — DaughtersCrown.jpeg

Exif Tool was run against the attachment to extract embedded metadata. Key findings are summarised below:

Metadata Field	Value
File Name	DaughtersCrown.jpeg
File Size	19 kB

Metadata Field	Value
File Type	JPEG
MIME Type	image/jpeg
Image Dimensions	822 x 435 pixels
Resolution	120 x 120 DPI
JFIF Version	1.01
Encoding	Baseline DCT, Huffman coding
File Creation Date	2021-01-25 16:41:00 +05:30
Zone Identifier	Exists (downloaded from internet)

The Zone Identifier field confirms the file was downloaded from the internet. The file creation timestamp of 2021-01-25 aligns with the email send date, suggesting the attachment was prepared immediately before the campaign launch.

5.4 OSINT Investigation — emkei.cz

Open-source intelligence (OSINT) was conducted on the relay server emkei.cz. A Google search confirmed that emkei.cz is a publicly accessible Fake Mailer service — allowing anyone to send emails with any From address, no authentication required, operational since 2009.

OSINT Field	Finding
Service Name	Emkei's Fake Mailer
URL	https://emkei.cz
IP Address	93.99.104.210
Operational Since	2009
Nature	Anonymous fake email service — no sender authentication
Threat Actor Use	Abused for phishing, BEC, and social engineering campaigns
Header Fingerprint	Received: from localhost (emkei.cz [93.99.104.210])
Threat Level	HIGH — known malicious infrastructure

5.5 Hidden Content — Excel Attachment (Sheet3)

Analysis of an Excel attachment found within the email revealed a hidden sheet (Sheet3) using black background with white text — a concealment technique designed to hide content from casual inspection. The hidden cells contained Base64-encoded strings which upon decoding revealed:

The Martian Colony, Beside Interplanetary Spaceport.

This decoded value directly corresponds to the fictional location in the email body lore, confirming the attacker embedded encoded clues consistently across both the email and hidden attachment data.

Finding	Detail
Sheet Name	Sheet3
Concealment Method	White text on black background — invisible to casual viewer
Encoding	Base64
Decoded Value	The Martian Colony, Beside Interplanetary Spaceport.
Analyst Action	Select all cells, change font colour to black to reveal hidden text
Significance	Attacker embedded encoded clues across multiple file types

6. Indicators of Compromise (IOCs)

IOC Type	Value	Context
IP Address	93.99.104.210	<i>emkei.cz relay server — malicious sender infrastructure</i>
Domain	emkei.cz	<i>Anonymous fake email sender service abused by attacker</i>
Domain	microapple.com	<i>Typosquat domain impersonating Apple Inc.</i>
Email Address	billjobs@microapple.com	<i>Spoofed sender address</i>
Email Address	negeja3921@pashter.com	<i>Attacker-controlled Reply-To inbox</i>
Domain	pashter.com	<i>Disposable email service hosting attacker inbox</i>
File	DaughtersCrown.jpeg	<i>Attachment — JPEG, 19kB, Zone ID exists</i>
File	GoodJobMajor.pdf	<i>Attachment — PDF, magic bytes 25 50 44 46</i>

7. Tools & Techniques Used

- Notepad++ — Raw .eml file inspection and header extraction
- CyberChef — Base64 decoding and hex conversion of encoded email body
- filesig.search.org — File signature (magic bytes) lookup and verification
- ExifTool — Metadata extraction from JPEG and PDF attachments
- Manual Header Analysis — Field-by-field inspection of SMTP routing, authentication, and spoofing indicators

8. Recommendations

8.1 Immediate Actions

- Block IP 93.99.104.210 and domain emkei.cz at the email gateway and firewall
- Block sender domain microapple.com as a known typosquat
- Add negeja3921@pashter.com and billjobs@microapple.com to threat intelligence blocklists
- Submit IOCs to VirusTotal and internal SIEM for retroactive hunting

8.2 Detection Engineering

- Create SIEM rule to alert on emails where Reply-To differs from From address
- Create rule to flag emails originating from emkei.cz or similar anonymous relay services
- Alert on SPF FAIL + DKIM absent + no DMARC — triple authentication failure pattern
- Use Message-ID containing 'localhost' as an indicator of fake mailer usage

8.3 User Awareness

- Train users to verify Reply-To address before responding to any email
- Educate staff on typosquatting — microapple.com vs apple.com distinction
- Implement a mail client banner warning for external emails with mismatched Reply-To

9. Final Verdict

VERDICT: MALICIOUS — PHISHING EMAIL

The email "A Hope to CoCanDa" is a confirmed phishing email. The following factors collectively establish this verdict with high confidence:

- SPF authentication FAIL — sending IP not authorised for the claimed domain
- Sender relayed through emkei.cz — a known anonymous fake email service
- Reply-To mismatch — attacker redirecting victim replies to a controlled inbox
- Typosquat sender domain (microapple.com) impersonating Apple Inc.
- Base64-encoded body with social engineering lure content
- Attachments delivered with zone identifier confirming internet origin

— Mohamed Sikkanthar Asaf | Cyber Security Professional